

CPE 301 – Embedded Systems Design

Final Project Part 3

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Greenhouse Plant Monitoring Tank System

1.0 Project Motivation / Introduction

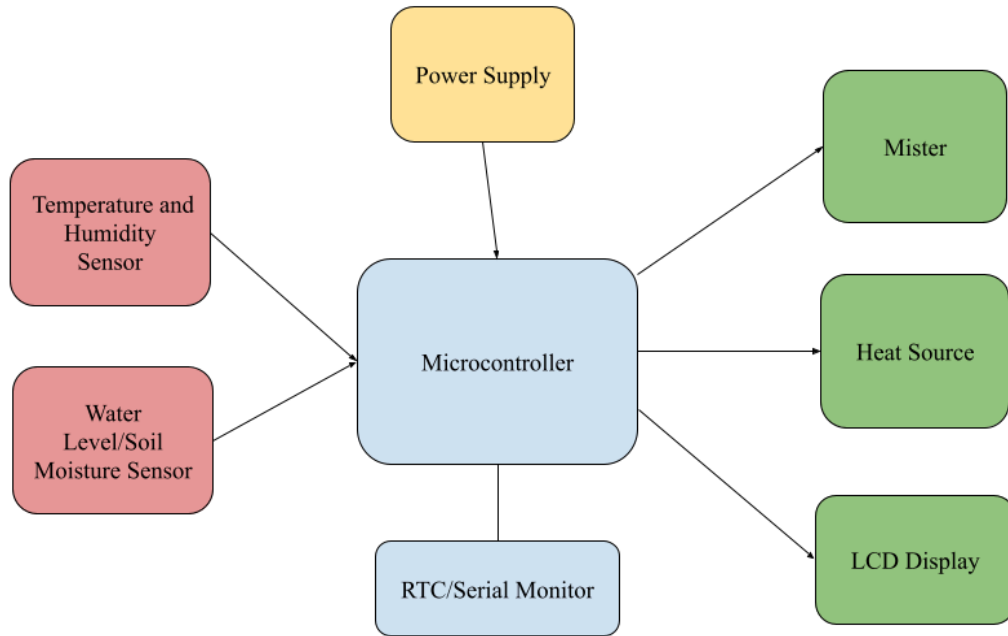
The Smart Plant Monitoring Tank System is designed to reduce the need for constant human monitoring of plant conditions. Plant health depends on temperature and humidity; improper conditions can lead to poor growth or plant death.

This project expands that idea by converting a fish tank into a small greenhouse that not only monitors but also controls the environment. A DHT11 sensor measures temperature and humidity, and the system automatically adjusts conditions using a heat source and a misting system. This allows plants to grow in optimal environmental conditions with minimal user effort.

2.0 Proposed System

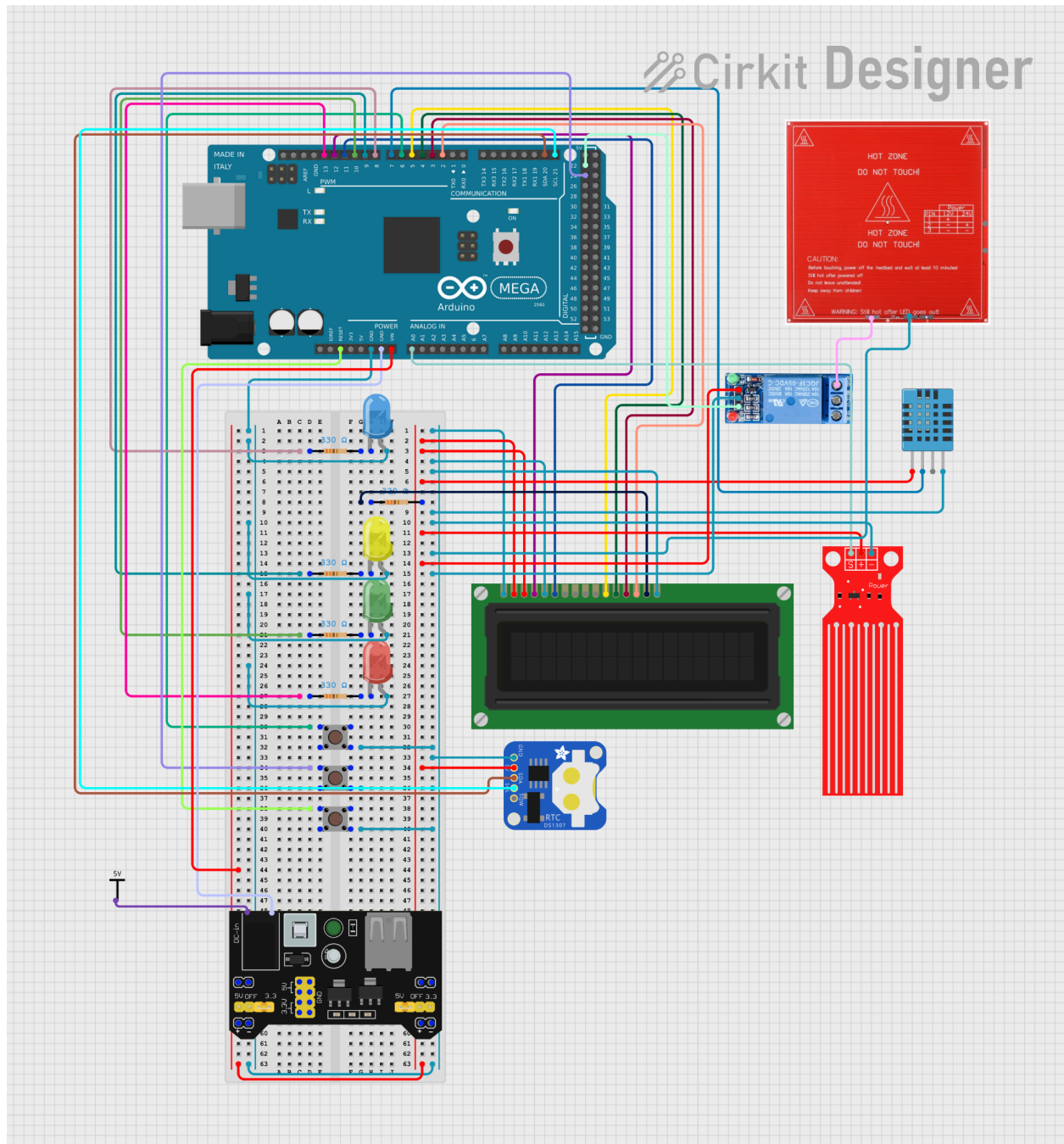
The system uses an Arduino Mega2560 to monitor and control environmental conditions inside the tank. The DHT11 sensor provides real-time temperature and humidity data, which the system compares to predefined thresholds. If conditions are outside the desired range, the system activates a heat source or mister to correct them.

2.1.1 System Block Diagram



The Arduino acts as the central controller, receiving input from sensors and sending output to actuators, displays, and LEDs. The system also includes buttons for control and an RTC module for time-based logging.

2.1.2 System Schematic Diagram



2.2 Component List

- DHT11 sensor tracks humidity and temperature.
 - docs.cirkitdesigner.com/component/a76bb123-3d3c-417d-b1f3-c8417efb7bc7
- Heat source pads provide heat when given power..
 - docs.cirkitdesigner.com/component/04df1491-52bf-41fb-96b8-b7d8282b21c6

- *Ultrasonic mist maker allows for humidity to be automatically controlled when provided power.*
 - docs.cirkitdesigner.com/component/a0726024-2717-468d-bcb0-1ef57233b392
 - *Mister couldn't be powered without additional components that wouldn't arrive in time, and thus we scrapped the implementation*
- LCD 16-pin display allows for displaying text and symbols on a readable screen.
 - docs.cirkitdesigner.com/component/23598c1c-7607-4538-b177-9ab948fb1937
- LED indicators for system state, light up when provided power
 - docs.cirkitdesigner.com/component/1310de1e-180e-4aed-a1c6-f80032a2bfde.
- RTC module synchronizes with real-world time, providing an accurate clock for events.
 - docs.cirkitdesigner.com/component/9f0127e3-b141-4b6c-b341-c271412e4837
- The water sensor reads water levels in the tank at all times, sending the data to the Arduino.
 - docs.cirkitdesigner.com/component/a7130bc6-78e1-2b0a-9c4e-1fe381aeb71f
- Button sends a signal when pressed, requires power at all times.
 - docs.cirkitdesigner.com/component/232ac546-6e45-4194-9a27-0fa614949a21
- Power supply, supplies power to the Arduino and circuit from an outlet, sending out 5V.
 - docs.cirkitdesigner.com/component/b73263be-92fa-4f42-99e3-f76b69d6f495
- Relay module allows for turning on and off a device by providing power and cutting it.
 - docs.cirkitdesigner.com/component/dcaa953f-76ee-47e1-bb67-b445bc4cfe37

3.0 System Behavior

When the system is turned ON, it initializes all components and begins monitoring the environment in the IDLE state. When turned OFF, all sensing and actuator activity stops. During operation, the system responds to environmental changes by activating or deactivating actuators as needed.

3.1 Module Design

Temperature and Humidity Sensor (DHT11)—Reads the temperature and humidity inside the tank and stores the values in variables. These values are continuously sent to the Control Logic module for decision-making.

Analog Sensor (e.g., Soil Moisture Sensor)—Reads an analog value (ADC) from the environment and stores it in a variable. This value can also be used by the Control Logic module to influence system behavior.

Control Logic—Decides whether the heat source or mister should be turned on or off based on the temperature and humidity levels. It also determines the current system state (OFF, IDLE, ACTIVE, ERROR) and sends signals to the actuator, display, LED, and communication modules.

Heat Source (Actuator)—Enabled or disabled based on output from the Control Logic module, increasing the temperature inside the tank when it is too low.

Mister (Actuator)—Enabled or disabled based on output from the Control Logic module, increasing humidity when it falls below the desired level.

LCD Display—Reads temperature, humidity, and system state values from the Control Logic module, displaying them to the user in real time.

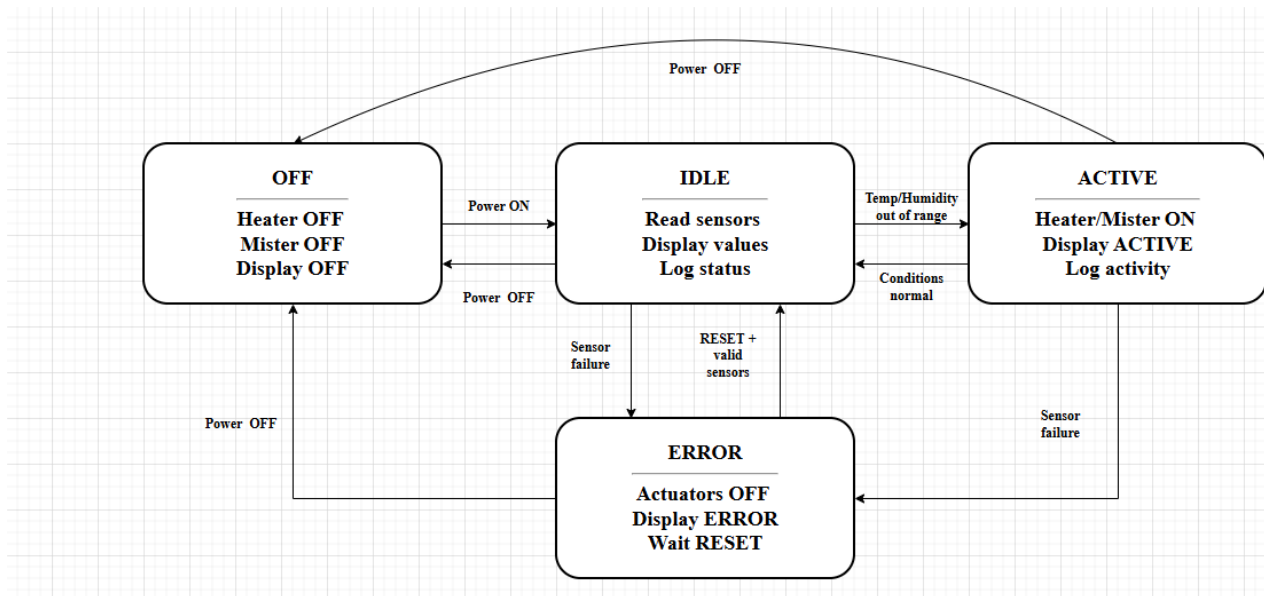
LED Indicators—Receive signals from the Control Logic module and indicate the current system state (OFF, IDLE, ACTIVE, or ERROR), with only one LED active at a time.

RTC + Serial Communication Module—Receives system events from the Control Logic module and logs state changes, actuator activity, and errors with timestamps to the Serial Monitor.

The user interacts with the system using three buttons: ON, OFF, and RESET. The ON button (interrupt-based) starts the system, the OFF button shuts it down from any state, and the RESET button clears errors and returns the system to IDLE. LEDs and the LCD provide feedback to the user.

The system continuously monitors temperature and humidity and reacts accordingly. If values fall outside the desired range, the system enters ACTIVE mode and turns on the appropriate actuator. If conditions return to normal, the system goes back to monitoring. If a sensor fault occurs, the system enters ERROR mode and shuts down actuators.

3.2 Control Logic (State Diagram)



The system operates using four states: OFF, IDLE, ACTIVE, and ERROR. In OFF, the system is inactive. In IDLE, it monitors conditions. In ACTIVE, it turns on actuators to correct conditions. In ERROR, it shuts down actuators due to sensor faults and waits for RESET.

LED Mapping

State	LED
OFF	LED1

IDLE LED2

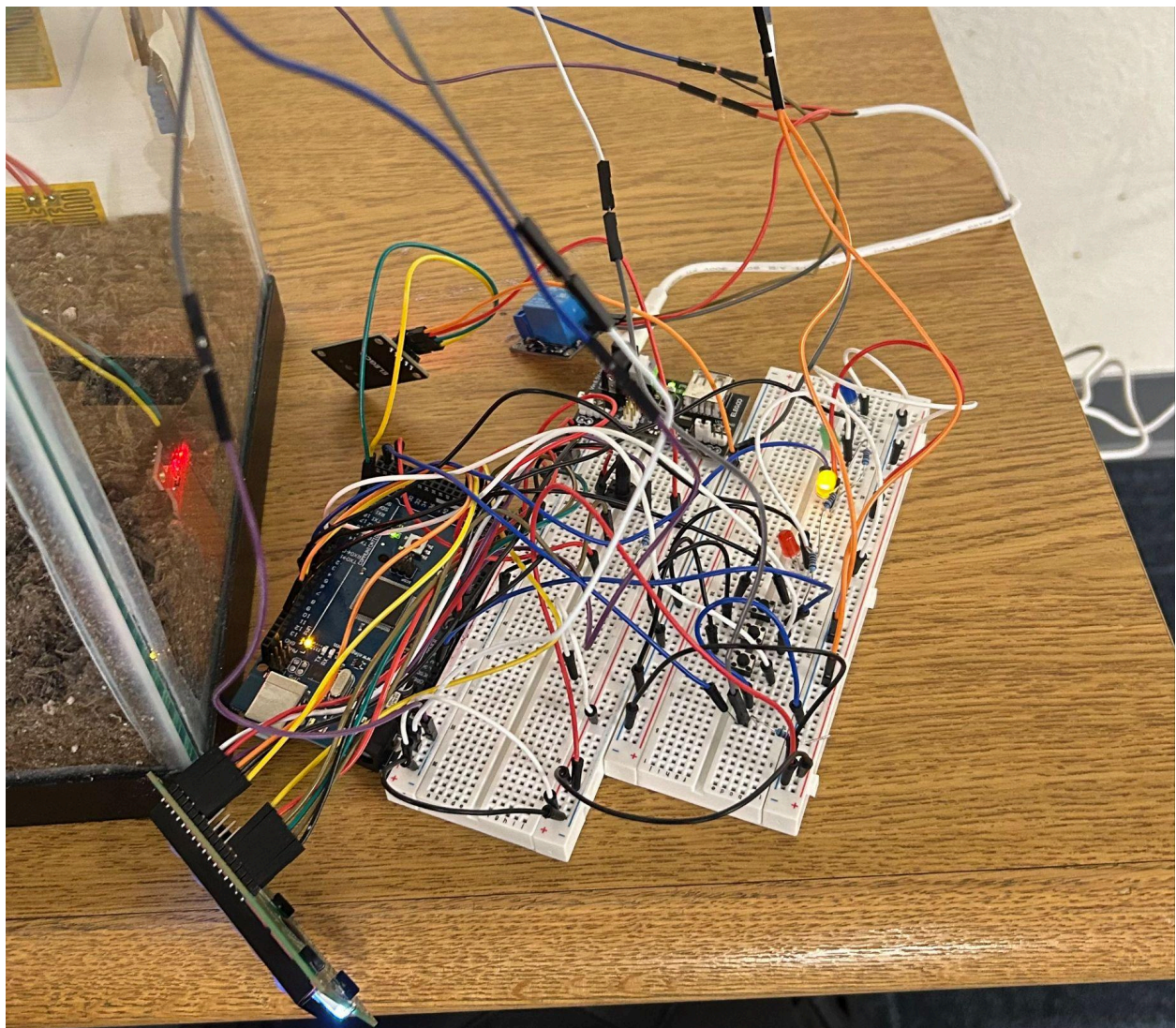
ACTIVE LED3

ERROR LED4

4.0 Results

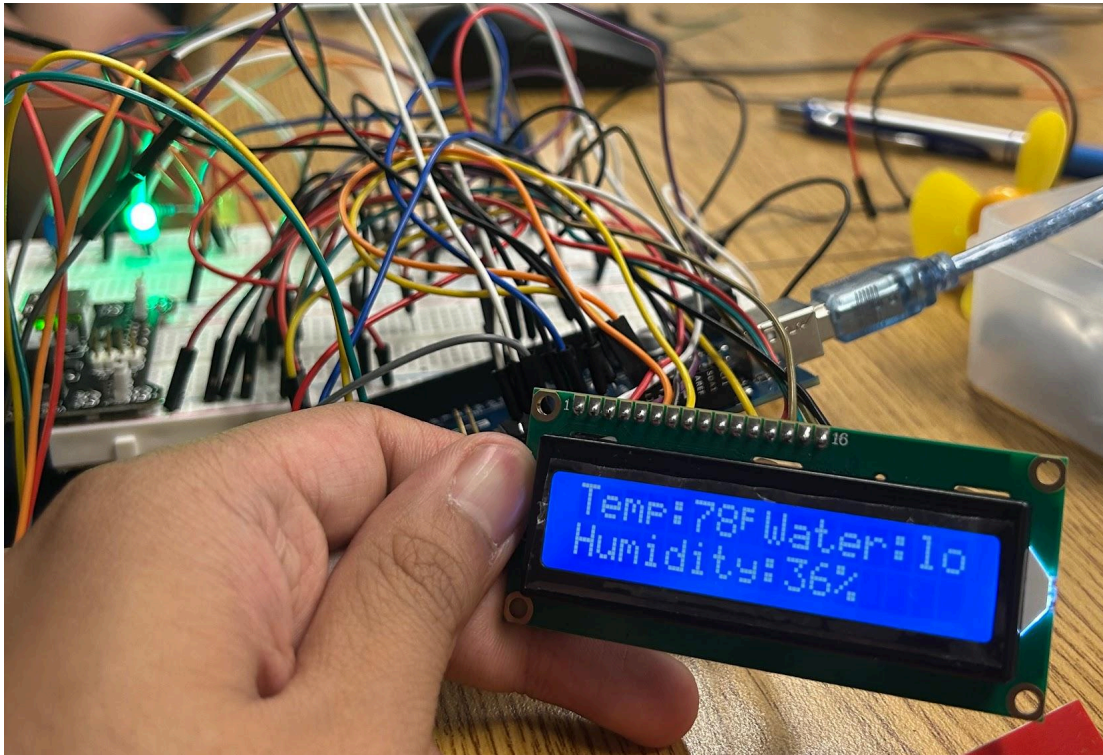
NOTE: For demonstration purposes, the circuit would update every five seconds instead of one minute. Additionally, the temperature threshold to turn on the heat actuator was changed from 50 degrees to 80 degrees.

4.1 Circuit

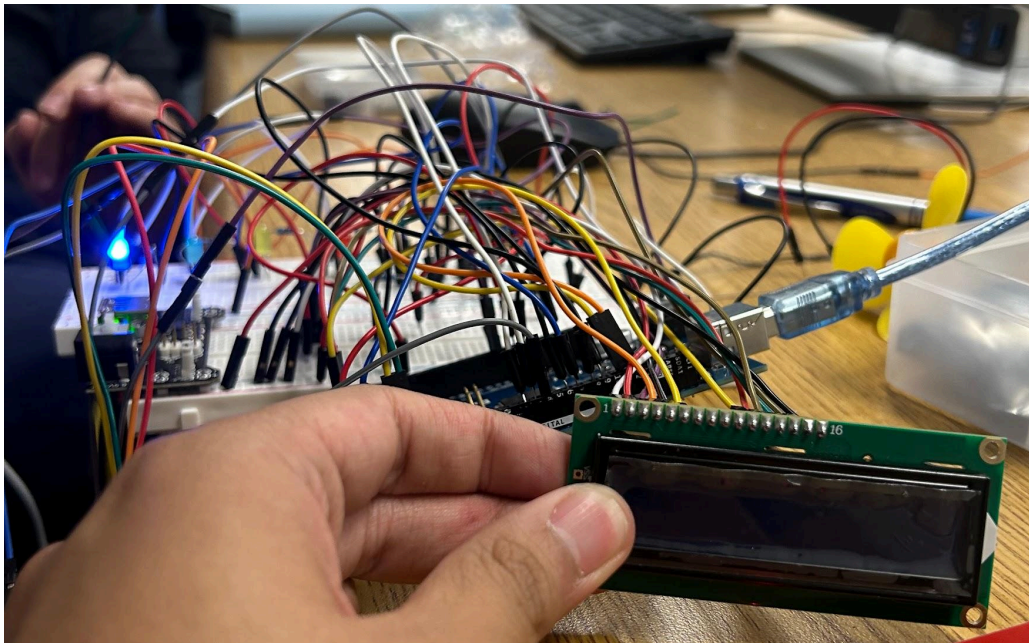


4.2 State Images

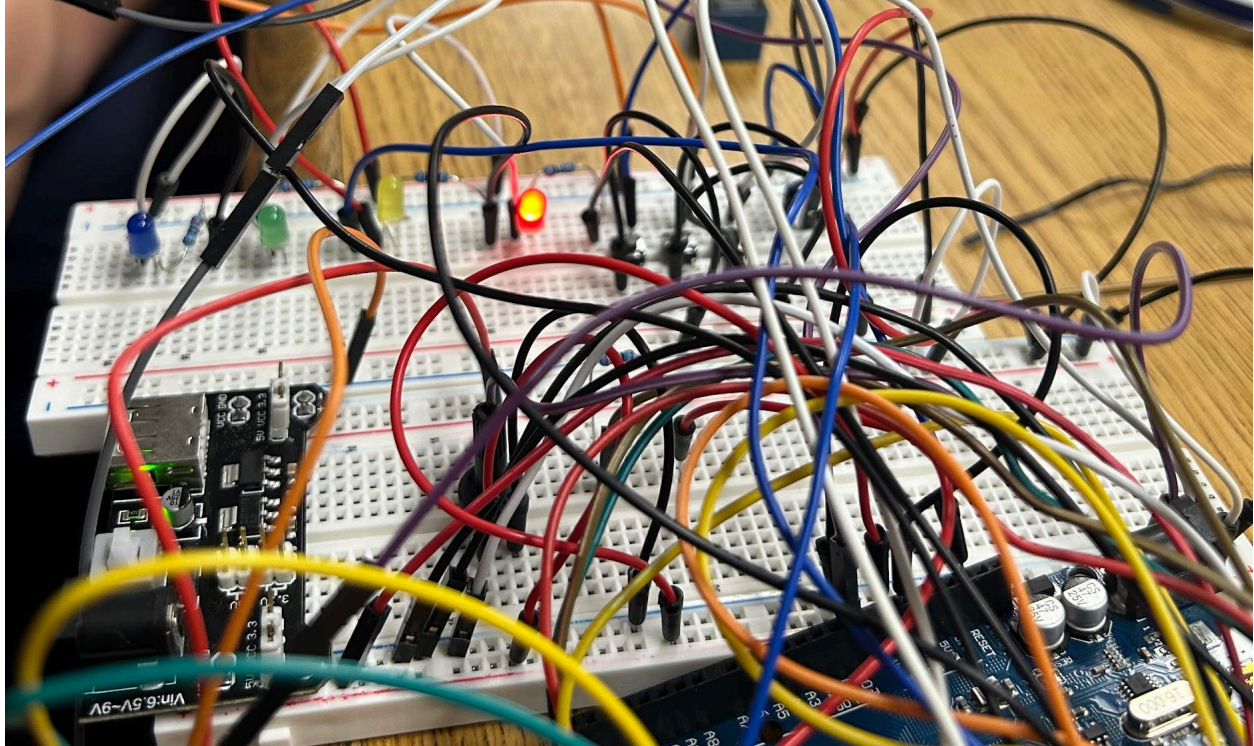
Green - ACTIVE



Blue - OFF



Red - ERROR



Yellow - IDLE



4.3 Serial Monitor

Output Serial Monitor X

Message (Enter to send message to 'Arduino Mega or Mega 2560' on 'COM9')

```
PIN SETUP WAS FINISHED
OFF button pressed
ON button pressed
2026/5/12 7:46:32      Temperature: 76 degrees Humidity: 22%   Water Level: 12 (Low)
2026/5/12 7:46:37      Temperature: 76 degrees Humidity: 22%   Water Level: 13 (Low)
2026/5/12 7:46:42      Temperature: 76 degrees Humidity: 22%   Water Level: 23 (Low)
2026/5/12 7:46:47      Temperature: 76 degrees Humidity: 22%   Water Level: 35 (Low)
2026/5/12 7:46:52      Temperature: 76 degrees Humidity: 22%   Water Level: 141 (Low)
2026/5/12 7:46:57      Temperature: 76 degrees Humidity: 22%   Water Level: 188 (Low)
2026/5/12 7:47:2       Temperature: 76 degrees Humidity: 22%   Water Level: 213 (High)
2026/5/12 7:47:7       Temperature: 76 degrees Humidity: 22%   Water Level: 13 (Low)
2026/5/12 7:47:12      Temperature: 76 degrees Humidity: 22%   Water Level: 11 (Low)
OFF button pressed
ON button pressed
2026/5/12 7:48:50      Temperature: 76 degrees Humidity: 22%   WaterON button pressed
ON button pressed
Level: 9 (Low)
2026/5/12 7:48:55      Temperature: 76 degrees Humidity: 22%   Water Level: 8 (Low)
2026/5/12 7:49:0       Temperature: 76 degrees Humidity: 22%   Water Level: 9 (Low)
2026/5/12 7:49:5       Temperature: 76 degrees Humidity: 22%   Water Level: 9 (Low)
2026/5/12 7:49:10      Temperature: 76 degrees Humidity: 22%   Water Level: 9 (Low)
2026/5/12 7:49:15      Temperature: 76 degrees Humidity: 22%   Water Level: 9 (Low)
2026/5/12 7:49:20      Temperature: 76 degrees Humidity: 22%   Water Level: 9 (Low)
OFF button pressed
```

5.0 Environmental & Energy Impact

Energy Efficiency

The system minimizes energy consumption by only activating actuators when environmental conditions fall outside the desired range. During normal operation in the IDLE state, only low-power components such as the sensor and display are active, reducing unnecessary energy usage.

Design Safety

All components operate within the voltage and current limits of the Arduino Mega2560. Actuators such as the heat source and mister are controlled using proper interfacing components (e.g., transistors or relays) to prevent damage. Additionally, the system includes an ERROR state that immediately turns off actuators if a sensor fault is detected, ensuring safe operation.

Affordability

The system is designed using mostly components provided in the course kit, such as the Arduino, LEDs, and sensors. Additional components like the tank, heat source, and mister are low-cost and easily accessible, making the project affordable to build and replicate.

Sustainability

The system promotes sustainability by reducing energy and water waste through controlled activation of actuators. By maintaining optimal conditions automatically, it also helps prevent plant loss, reducing resource waste over time.

Accessibility

The system is designed to be simple and user-friendly, with clear LED indicators, an LCD display, and basic button controls. This allows users with limited technical experience to easily operate and understand the system.

6.0 Conclusion

6.1 System Functionality

The system mostly operated as intended, with all states functioning correctly throughout testing. Upon startup, the device initializes in an OFF mode; pressing the power button transitions the system into an ON mode. In the ON mode, the system enters IDLE state, where sensor data is continuously monitored. When the sensor readings fall within specified thresholds, the system transitions into the ACTIVE state, activating components such as the heat pads. If a fault is detected, the system enters the ERROR mode, which can be cleared by pressing the reset button.

6.2 Limitations

The purchased mister component required a separate atomization circuit board to operate and control the mister frequencies. Since this circuit board was not included with the component, the mister could not be successfully integrated into the system.

6.3 Video Link

[Smart Plant Monitoring Tank System Video](#) (also in GitHub)